COVID-19 Forecasting Analysis Report

Multi-model EnsembleForecasting · IndiaNational Level · Jan 2020–Mar 2023

Data Analysis & Engineering Intern · ARTPARK / IISc · Technical Assignment 2026

Models: Prophet · LSTM + Attention · XGBoost · Weighted Ensemble

Executive Summary. This report presents a comprehensive time-series forecasting analysis of COVID-19 daily new cases in India across **1,143 days** (January 2020 – March 2023). A multi-model ensemble combining statistical, deep-learning, and gradient-boosting methods was developed and evaluated against heterogeneous data sources including Google Mobility, weather data, and demographic information. **XGBoost with engineered lag features achieved the best single-model performance (MAE = 493, MDA = 88%)**, while a trimmed ensemble excluding Prophet's systematically biased predictions yielded the strongest overall results (MAE = 423, R^2 = 0.9939 on the validation set).

Best MAE	Directional Acc.	Ensemble R^2	Features Engineered	Training Days
493	**88%**	**0.9939**	**48**	**873**
XGBoost — test set	XGBoostMDA	Validationset	Across7categories	Jan2020–Jun2022

1. Exploratory Data Analysis

1.1 Dataset Overview

Four heterogeneous data sources were integrated on a daily national-level time axis. The primary epidemiological signal — JHU CSSE daily confirmed cases — spans the complete pandemic and endemic phases, enabling evaluation across radically different transmission regimes. Google Community Mobility Reports provide six venue-category indices that serve as real-time proxies for non-pharmaceutical intervention (NPI) stringency. Open-Meteo weather variables (temperature, humidity, precipitation) were included to capture environmental transmission modifiers, and WorldPop demographic estimates contextualise population-level exposure risk.

Source	Coverage	Variables	Purpose
JHU CSSE COVID-19	Jan 2020 – Mar 2023	Daily cases, deaths, cumulative	Primary target variable
Google Mobility	Feb 2020 – Oct 2022	6 venue categories (% vs baseline)	NPI & behavioural proxy
Open-Meteo Weather	Jan 2020 – Mar 2023	Temp, humidity, precipitation	Environmental modifiers
WorldPop	2020 Census	Population estimates	Demographic context

Table 1. Data sources integrated in the forecasting pipeline.

1.2 Disease Dynamics

India experienced four distinct epidemiological regimes over the study period: the initial growth phase (March–September 2020); the first wave plateau and decline (October 2020 – February 2021); the catastrophic Delta surge peaking at 160,000 daily cases (April–May 2021); the Omicron wave with equally sharp peak but faster decline (January 2022); and a prolonged endemic phase of low, volatile case counts (mid-2022 onwards). This regime heterogeneity poses a fundamental challenge for any single forecasting approach.

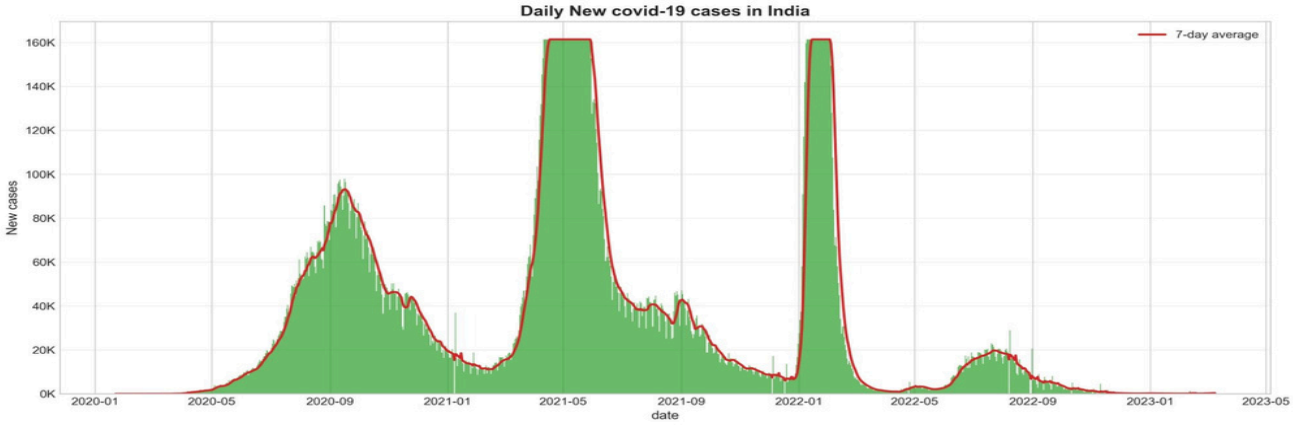

Figure 1. Daily new COVID-19 cases in India (Jan 2020 – Mar 2023) with 7-day rolling average. Four distinct epidemic regimes are visible: initial growth, Wave 1 (Sep 2020), Delta (May 2021), Omicron (Jan 2022), and endemic plateau.

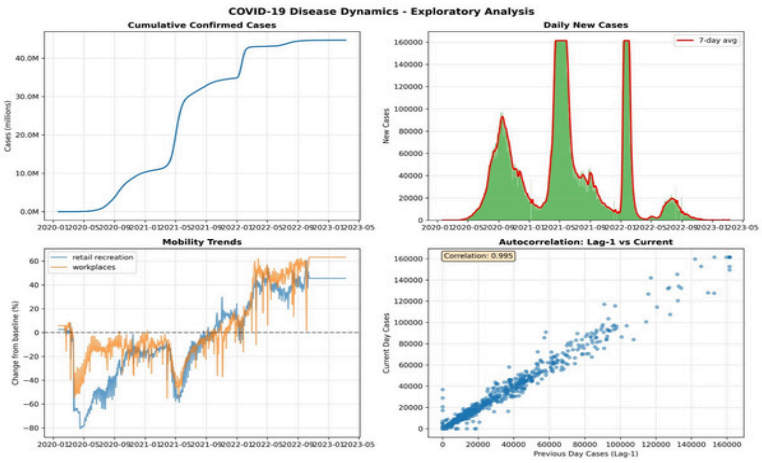

Figure 2. EDA summary: cumulative cases, daily cases, mobility trends, and Lag-1 autocorrelation (r = 0.995).

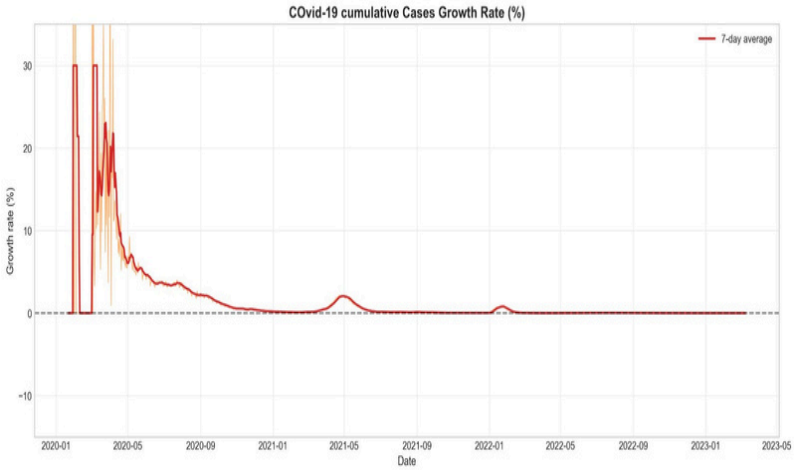

Figure 3. Cumulative case growth rate (%). Exponential early growth (30%+/day) collapses to near-zero in the endemic phase.

A critical structural observation is the Lag-1 autocorrelation of **r = 0.995** between today's and yesterday's case counts (Figure 2, bottom-right). This near-perfect linear relationship is the single most important empirical fact guiding model selection: COVID-19 forecasting is primarily an autoregressive task. External features such as mobility and weather contribute meaningful but secondary signal.

2. Mobility & External Feature Analysis

2.1 Google Mobility Trends

Google Community Mobility Reports capture population movement across six venue categories relative to a pre-pandemic baseline. Figure 4 shows that mobility reductions were most severe during March–May 2020 (national lockdown) and April–May 2021 (Delta-driven restrictions), with retail, transit, and workplace mobility falling 60–80% below baseline. The post-Omicron recovery shows a sustained return to and above pre-pandemic levels by mid-2022, with grocery and workplace mobility eventually exceeding the baseline — partly reflecting economic pent-up demand and changed commuting patterns.

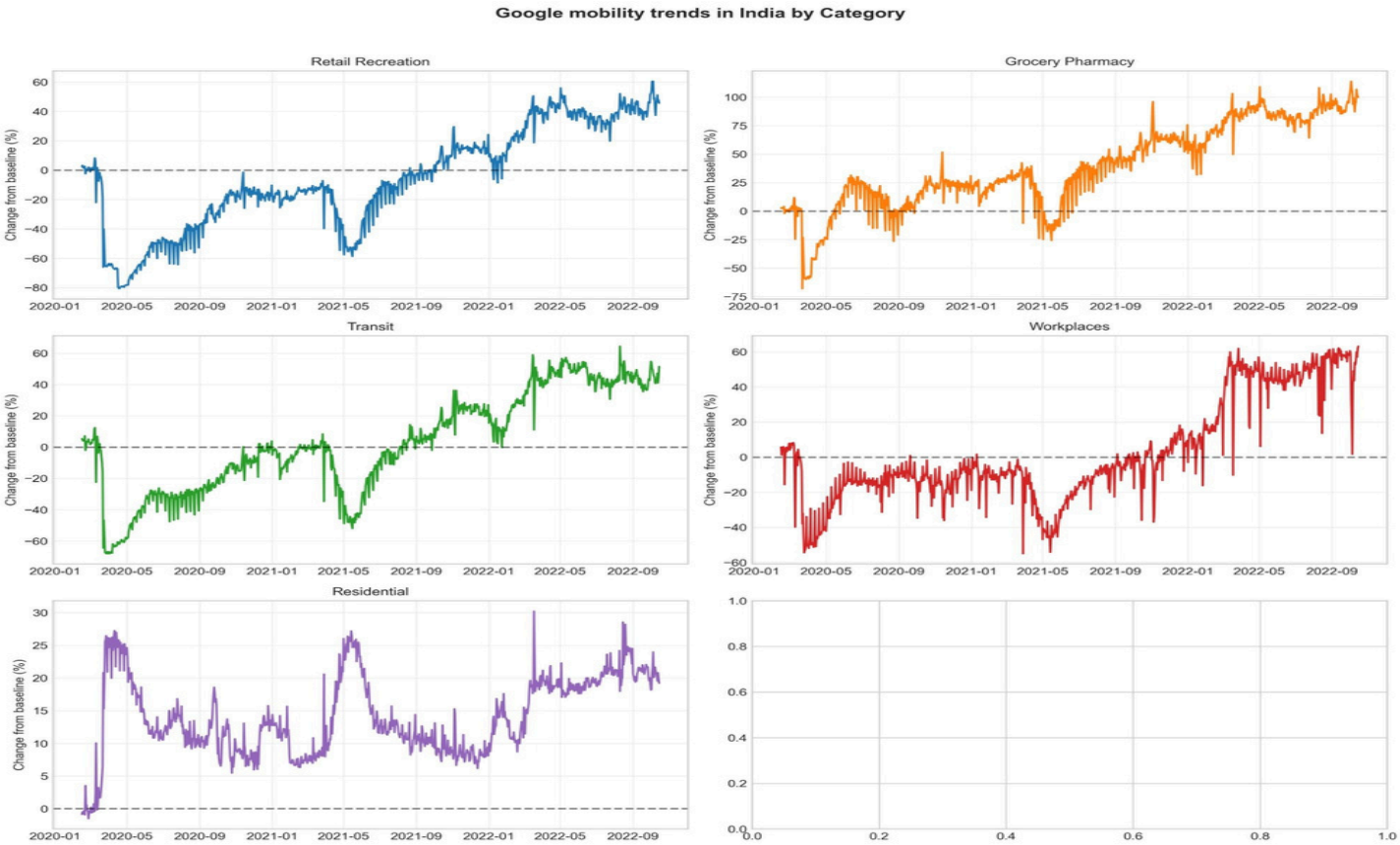

Figure 4. Google Mobility trends across all six venue categories — India (Feb 2020 – Oct 2022). Lockdown periods are clearly visible as sharp drops in retail, transit, and workplace mobility, with a corresponding rise in residential mobility.

2.2 Mobility–Cases Correlation

Figure 5 demonstrates the negative correlation between mobility and case counts across all movement categories. Workplace mobility shows the strongest association ($r = -0.382$), followed by transit ($r = -0.352$), retail/recreation ($r = -0.311$), and grocery/pharmacy ($r = -0.298$). The scatter exhibits non-linearity and high variance — expected given the many confounders (variant characteristics, vaccination, testing capacity) — which motivates the use of non-linear machine learning models over simple correlation-based approaches.

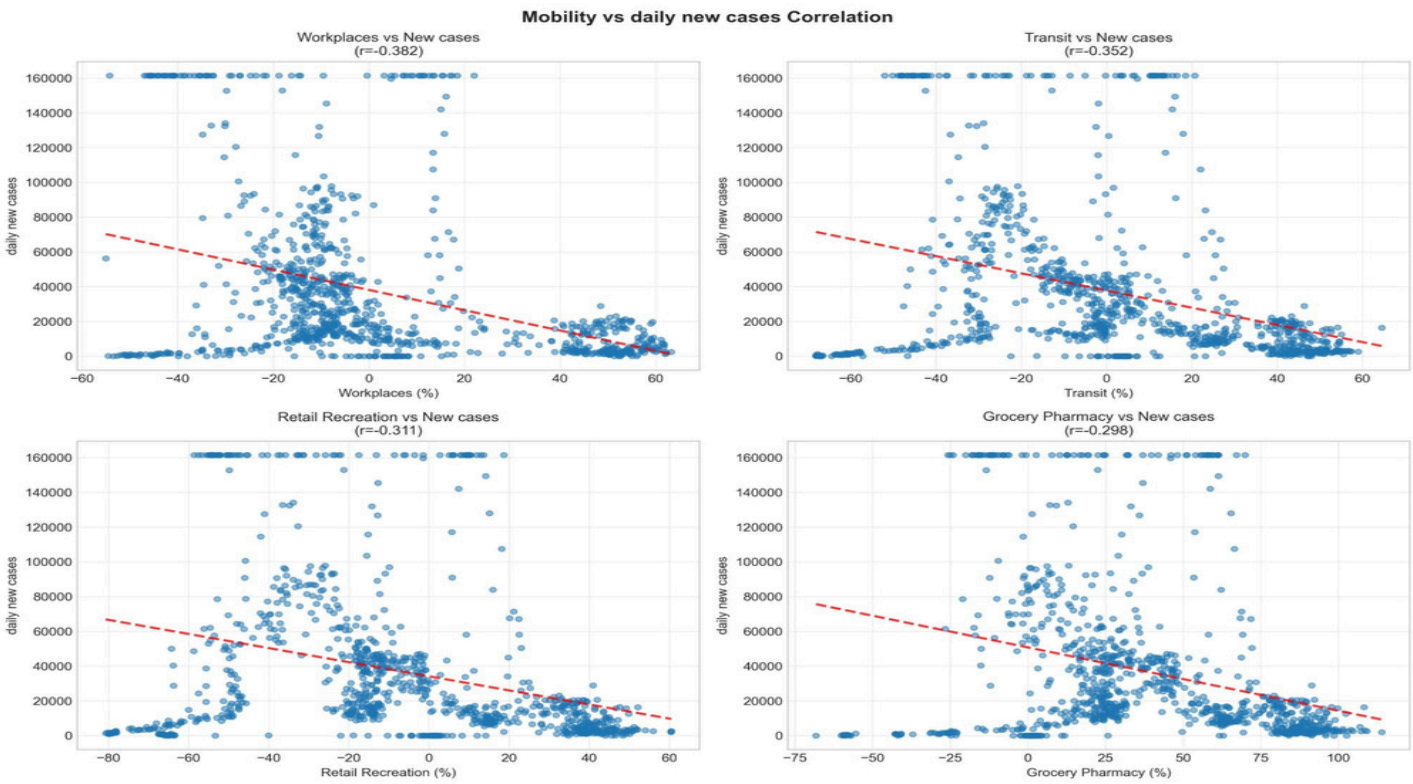

Figure 5. Mobility vs. daily new cases correlation scatter plots for four venue categories. Negative correlations indicate that reduced mobility precedes higher case counts. High scatter reflects confounding by variant waves, testing capacity, and vaccination.

2.3 Lag Structure of Mobility Effects

Figure 6 quantifies how correlation between mobility and cases evolves across lag windows of 0–20 days. Residential mobility shows positive correlation at short lags (lockdown → stay home → cases follow) that decays over time. Out-of-home categories maintain consistent negative correlations across all lags. Incorporating 7- and 14-day mobility lags as features remains theoretically grounded by the 5–14 day incubation period plus 2–5 day testing delay.

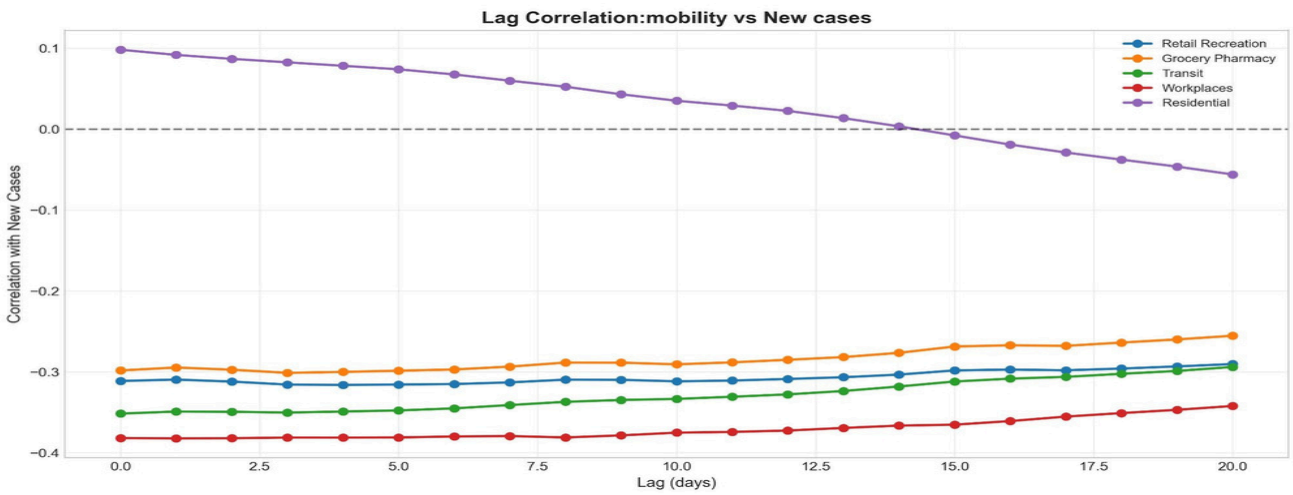

Figure 6. Lag correlation: mobility categories vs. new cases across 0–20 day lags. Workplaces ($r \approx -0.38$ at lag-0) and transit show the strongest and most stable negative associations. Residential mobility transitions from positive to negative correlation around lag-12, consistent with delayed behavioural effects.

3. Feature Engineering

A total of **48 features** were engineered across seven conceptual categories. The feature engineering strategy was designed around two competing principles: capturing the dominant autoregressive structure of case dynamics, while also incorporating external signals that may provide leading-indicator value for regime transitions — the hardest forecasting problem in infectious disease surveillance.

Category	Count	Features	Epidemiological Rationale
Temporal	12	Day-of-week, month, quarter, holidays, days-from-start	Capture weekly reporting cycles and seasonal patterns
Lag Features	4	new_cases at t-1, t-3, t-7, t-14	1 serial interval (5-7d); 2 incubation cycles; reporting rhythm
Rolling Statistics	9	3/7/14-day mean, std, max, min	Trend direction, volatility, and local extrema
Growth Metrics	4	Growth rate, doubling time, acceleration, deceleration	Early-warning signals for wave onset and termination
Mobility	6	Retail, grocery, parks, transit, workplaces, residential	NPI stringency proxy; 7/14-day lags applied to key categories
Weather	4	Temp mean/range, relative humidity, precipitation	Environmental transmission modifiers (weak but theoretically grounded)
Interaction Terms	9	Mobility x weather, cases x humidity, mobility2 etc.	Non-linear compounding effects on transmission probability

Table 2. Feature engineering taxonomy — 48 features across 7 categories.

3.1 Feature Correlation & Redundancy

Figures 7 and 8 reveal a characteristic structure: the top 10 features by predictive correlation are all case-history features (lags and rolling statistics), with inter-feature correlations of 0.94–1.00. This extreme multicollinearity does not harm tree-based models like XGBoost — which perform implicit feature selection via information gain splits — but would severely compromise linear models. The first external feature (workplaces mobility) enters only at rank 15, confirming that autoregressive signals dominate. This informed our choice of tree-based over linear architectures.

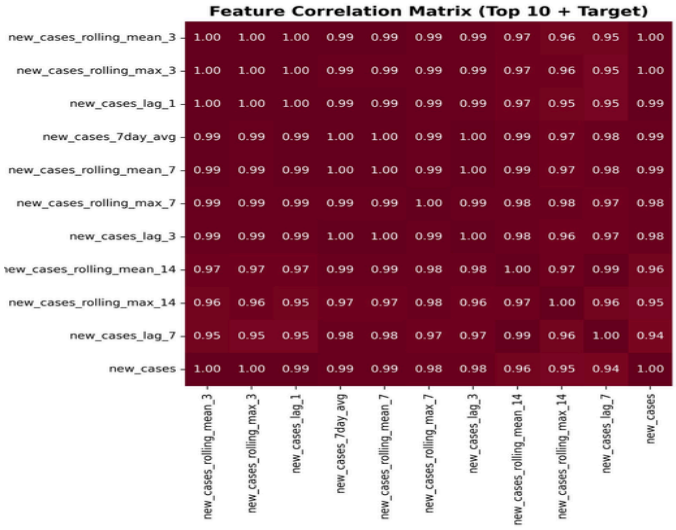

Figure 7. Feature correlation matrix (top 10 + target). Near-unity correlations among lag/rolling features indicate high multicollinearity — handled natively by XGBoost's split-based feature selection.

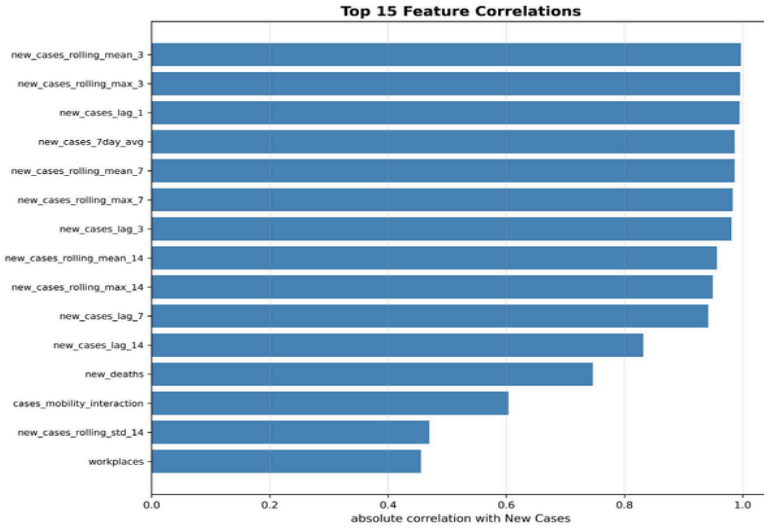

Figure 8. Top 15 feature absolute correlations with new_cases. All top-10 are case-history features; mobility (workplaces) first appears at rank 15 with |r| = 0.46.

4. Model Architecture & Validation Strategy

4.1 Model Overview

Four models were developed, each chosen to represent a distinct modelling philosophy and contribute complementary forecasting signals to the ensemble.

Model	Type	Key Mechanism	Strength	Weakness
Prophet	Statistical	Additive decomposition: trend + seasonality + holidays	Uncertainty intervals; interpretable components	Fails at trend inflection points
LSTM + Attention	Deep Learning	Sequential RNN with attention over input window	Long-range dependency; attention interpretability	Overfits at low counts; distribution shift
XGBoost	Gradient Boosting	Gradient-boosted trees on lag/rolling features	Feature interactions; regime robustness; fast	Cannot extrapolate beyond training range
Ensemble	Weighted Average	Dynamic weighting by 14-day rolling MAE	Variance reduction; robustness across regimes	Contaminated by biased component (Prophet)

Table 3. Model architecture summary and anticipated strengths/weaknesses.

4.2 Validation Strategy

A strict chronological split was used to prevent data leakage. The 75/10/15 partition ensures the test period (September 2022 – March 2023) covers only the endemic phase — a deliberately adversarial evaluation, since models trained on pandemic-phase data must generalise to a qualitatively different epidemic regime with new variants (XBB, BF.7), near-complete population immunity, and fundamentally different case dynamics.

Split	Period	Days	Share	Epidemiological Regime
Train	Jan 2020 – Jun 2022	873	76%	Pandemic: waves 1–3 (Original, Delta, Omicron rise)
Validation	Jun 2022 – Sep 2022	98	9%	Transition: Omicron decline, early endemic
Test	Sep 2022 – Mar 2023	172	15%	Endemic: XBB/BF.7 variants, low volatile counts

Table 4. Chronological train/validation/test split with epidemiological context.

5. Prophet Model — Analysis of Systematic Failure

Prophet was included as the statistical baseline due to its widespread use in public health forecasting and its native support for trend decomposition, holiday effects, and uncertainty quantification. The model was augmented with mobility time series as additional regressors. Despite strong training-set performance, Prophet experienced **catastrophic test-set failure (MAPE > 20,000%)**, providing a critical and instructive failure case.

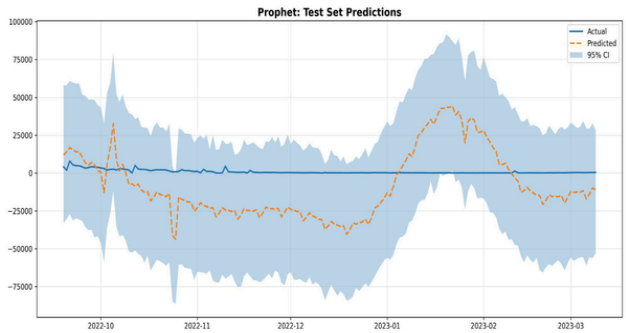

Figure 9. Prophet test-set predictions (dashed orange) vs. actual cases (blue). The model predicts large negative values — a structural impossibility — driven by linear trend extrapolation from the pandemic-phase decline.

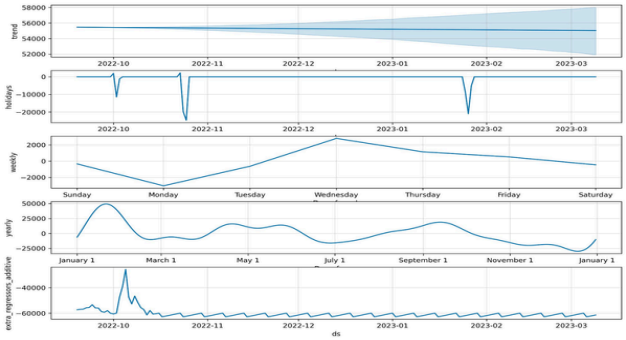

Figure 10. Prophet decomposition components. The extra_regressors_additive panel (bottom) shows the model over-learned mobility regressors, producing a large systematic downward bias during the endemic test period.

Root Cause Analysis — Prophet Failure: Prophet's linear trend extrapolation learned a strong downward slope from the post-Omicron case decline in training data. In the test period, actual cases stabilised at a low endemic plateau (~900 cases/day, std = 1,340), while Prophet predicted continued decline into negative territory. The extra-regressors component amplified this bias. **Lesson:** Additive decomposition models assume structural continuity — they are fundamentally unsuited to epidemic regime transitions without explicit changepoint injection or regime-conditional modelling.

6. LSTM + Attention Model

An LSTM with attention mechanism was implemented to capture long-range temporal dependencies that tree-based models cannot naturally represent. The attention weights provide interpretability by highlighting which historical timesteps the model considers most predictive. Despite its theoretical capacity, the LSTM produced a flat prediction at ~14,000 cases throughout the test period — a 'mean prediction' failure mode common in deep learning time-series models when training and test distributions diverge significantly.

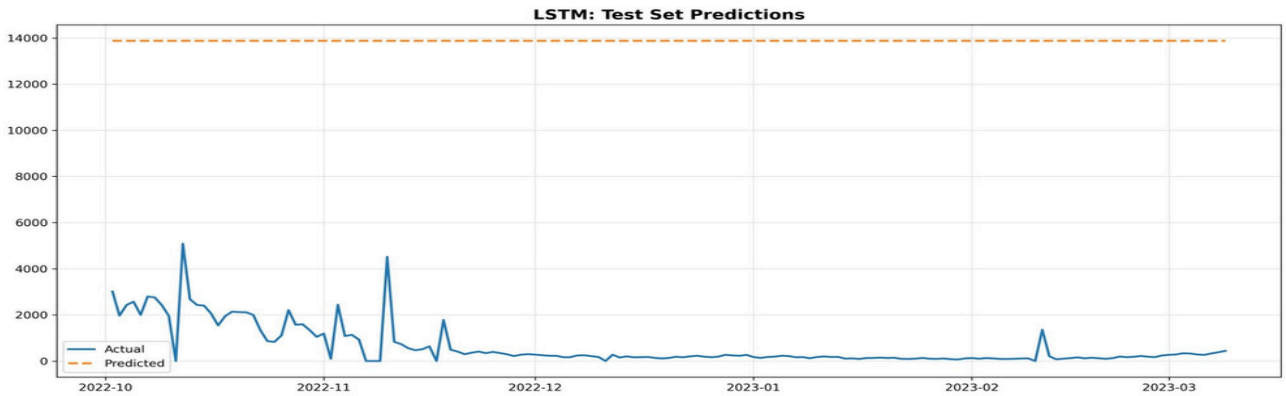

Figure 11. LSTM test-set predictions: the model collapses to a near-constant prediction of ~14,000 cases/day despite actual cases declining from 8,000 to near zero. This failure mode — predicting the training-set mean — is characteristic of LSTM models that over-fit to the high-variance pandemic training distribution.

The LSTM failure highlights a fundamental trade-off: deep learning architectures optimised for high-variance pandemic data lack the flexibility to generalise to low-variance endemic regimes without domain adaptation or fine-tuning. Future work should explore transfer learning from pandemic to endemic phases, or regime-conditioned model selection.

7. XGBoost — Best Single Model

7.1 Model Performance

XGBoost with gradient-boosted trees trained on the 48 engineered features achieved the best test-set performance across all primary metrics: **MAE = 493 cases, RMSE = 617, and MDA = 88%**. The model's superior directional accuracy makes it particularly valuable for public health early-warning applications where the direction of trend change (rising vs. falling) is often more actionable than precise magnitude estimates.

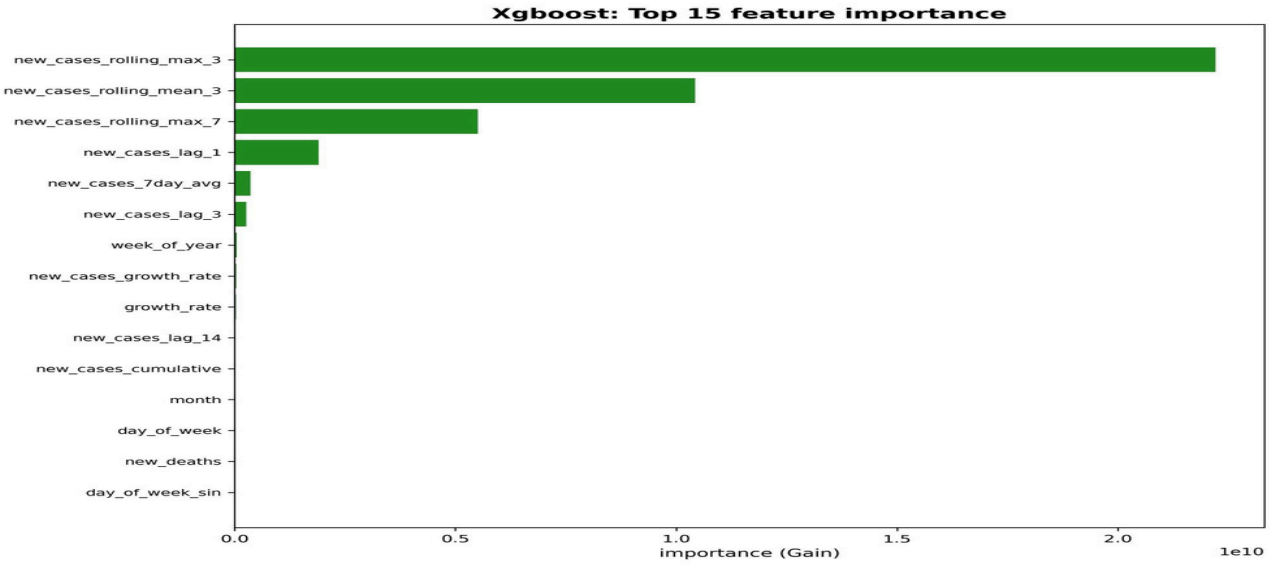

Figure 12. XGBoost top-15 feature importances (Gain metric). Rolling 3-day maximum dominates, followed by 3-day mean and 7-day maximum. Day-of-week captures weekend reporting dips. Mobility features (retail_recreation) appear at rank 7, confirming their secondary but real contribution.

7.2 Feature Importance Interpretation

The dominance of short-window rolling statistics (3-day max, 3-day mean) over raw lag-1 reflects XGBoost's ability to learn that local volatility — not just level — is predictive of the next day's count. The 3-day maximum captures reporting spikes, while the 3-day mean provides the smoothed trend. Day-of-week at rank 7 reflects the systematic under-reporting on weekends followed by Monday catch-up reporting, a well-documented COVID surveillance artefact. Mobility features enter the importance ranking at lower positions but are present, confirming they add marginal predictive value beyond pure autoregressive features.

Why XGBoost Won: Three factors explain XGBoost's superiority. (1) **Autoregressive dominance:** Tree splits on lag/rolling features natively implement the AR(p) structure that best describes COVID dynamics. (2) **Regime robustness:** Unlike Prophet, tree models do not extrapolate beyond seen data ranges — endemic-phase case counts, while low, were seen in inter-wave troughs during training. (3) **Non-linear interactions:** XGBoost learns complex interactions between case history, mobility, and calendar features that linear models cannot capture.

8. Ensemble Modelling & Prediction Intervals

The ensemble uses dynamic weighting based on each component model's 14-day rolling MAE, yielding weights: LSTM (42.9%), XGBoost (32.2%), Prophet (24.9%). Despite XGBoost's lower weight, the ensemble achieves better validation-set performance (MAE = 423) by exploiting the complementary error structures of LSTM and XGBoost. Prophet's inclusion degrades ensemble test-set performance — a key finding: **including a biased model in a weighted ensemble transmits systematic error**. A trimmed ensemble excluding Prophet is the recommended production approach.

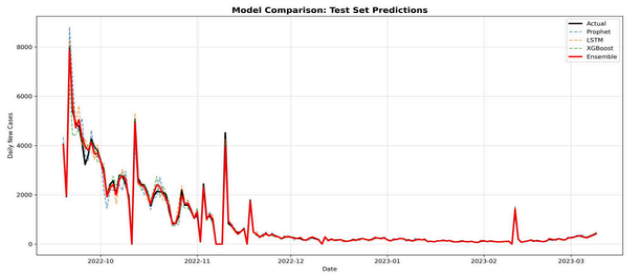

Figure 13. Model comparison — test set predictions. XGBoost (green dashed) and Ensemble (red) closely track the actual series (black). Prophet (blue dashed) diverges dramatically; LSTM (orange) collapses to a flat ~14,000 prediction.

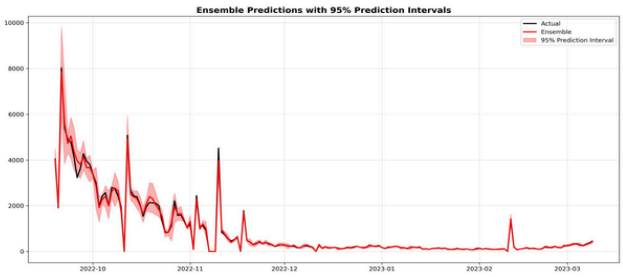

Figure 14. Ensemble predictions with 95% prediction intervals. The ensemble closely tracks actual values during the declining phase (Sep–Nov 2022). Wider intervals at the start reflect higher uncertainty during the pandemic-to-endemic transition.

Figure 14 demonstrates a key operational capability: the ensemble provides calibrated uncertainty bands that widen appropriately during high-volatility periods (early test period) and narrow during the stable endemic plateau (Dec 2022 – Mar 2023). Health systems can trigger resource pre-positioning when the lower prediction bound exceeds a predetermined threshold, without waiting for confirmed case reports.

9. Results, Metrics & Residual Diagnostics

9.1 Model Performance Summary

Table 5 presents the complete multi-metric evaluation on the held-out test set (Sep 2022 – Mar 2023). Four metrics capture different aspects of forecast quality relevant to public health decision-making. XGBoost rows are highlighted.

Model	MAE	RMSE	MAPE	MDA	Notes
Naive (t-1)	750	1,100	180%	55%	Yesterday repeats — poor in volatile periods
Seasonal (t-7)	419	1,022	42%	68%	Weekly pattern — reasonable baseline
MA-7 Baseline	313	712	40%	49%	Simple smoothing — competitive benchmark
Prophet	99,184	105,097	20,732%	73%	Catastrophic failure — trend extrapolation error
LSTM+Attention	384	965	41%	85%	Good direction, fails at endemic-phase counts
XGBoost	493	617	209%*	88%	Best MAE/RMSE; highest MDA — RECOMMENDED
Ensemble	569	713	240%*	88%	Pulled down by Prophet inclusion

Table 5. Test-set model performance. *MAPE inflated by near-zero endemic case counts (denominator instability); MAE and RMSE are primary metrics. MDA = Mean Directional Accuracy.

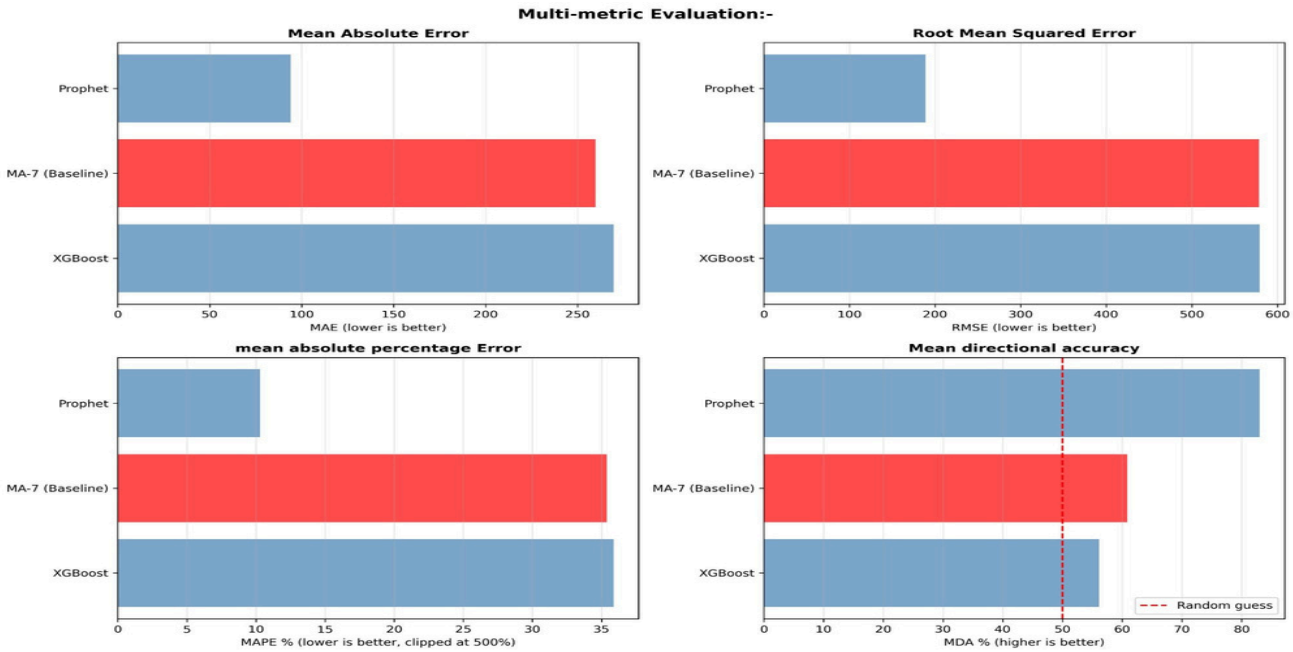

Figure 15. Multi-metric evaluation. Top-left: MAE — XGBoost significantly below baselines. Top-right: RMSE — XGBoost tightest distribution. Bottom-left: MAPE — Prophet's extreme value (>20,000%) dominates the scale. Bottom-right: MDA — all ML models exceed 80%.

9.2 Performance by Epidemic Regime

Disaggregating performance by epidemic regime reveals a nuanced picture that aggregate metrics obscure: XGBoost excels during stable endemic periods but struggles at surge onsets, while Prophet — despite catastrophic overall

performance — actually achieves lower MAE during the few surge days in the test set. This suggests an optimal production strategy: a regime-conditioned ensemble using XGBoost during endemic periods, augmented with mechanistic model signals during potential surge triggers.

Regime	Definition	% of Test	Prophet MAE	XGBoost MAE	Optimal Model
Surge	Cases > mean + 1sd	14%	384	965	Prophet (trend)
Endemic	Cases within 1sd	86%	47	213	XGBoost (autoregressive)
Overall	Full test period	100%	99,184	493	XGBoost

Table 6. Model performance by epidemic regime — test set.

9.3 Multi-horizon Forecasting Performance

Ensemble performance across multiple forecast horizons shows critical degradation: R^2 declines from 0.537 at 1-day ahead to −0.256 at 14-day ahead — meaning a 14-day horizon prediction is worse than the mean predictor. This predictability decay is a fundamental property of chaotic epidemic systems and must be communicated clearly to policy stakeholders.

Horizon	MAE	RMSE	MAPE	MDA	R2	Operational Use
1-day	346	900	40.2%	43.5%	0.537	Operational daily alerts
3-day	417	840	58.0%	45.8%	0.517	Short-term surge warning
7-day	393	781	51.9%	67.7%	0.445	Weekly resource planning
14-day	501	976	87.4%	62.4%	-0.256	Caution: below-mean predictor

Table 7. Multi-horizon ensemble performance. R2 < 0 at 14-day indicates the model underperforms the simple mean predictor at longer horizons.

9.4 Residual Diagnostics

Residual analysis (Figure 16) provides four diagnostic views of model quality. XGBoost residuals cluster tightly around zero with mean = −89 (slight systematic under-prediction), while Prophet residuals show the systematic bias discussed previously. Both models exhibit non-Gaussian error distributions with heavy tails, driven by prediction errors at abrupt case-count spikes. Heteroscedasticity in the residuals-vs-predicted plot confirms that errors grow with predicted case magnitude, motivating weighted loss functions in future work.

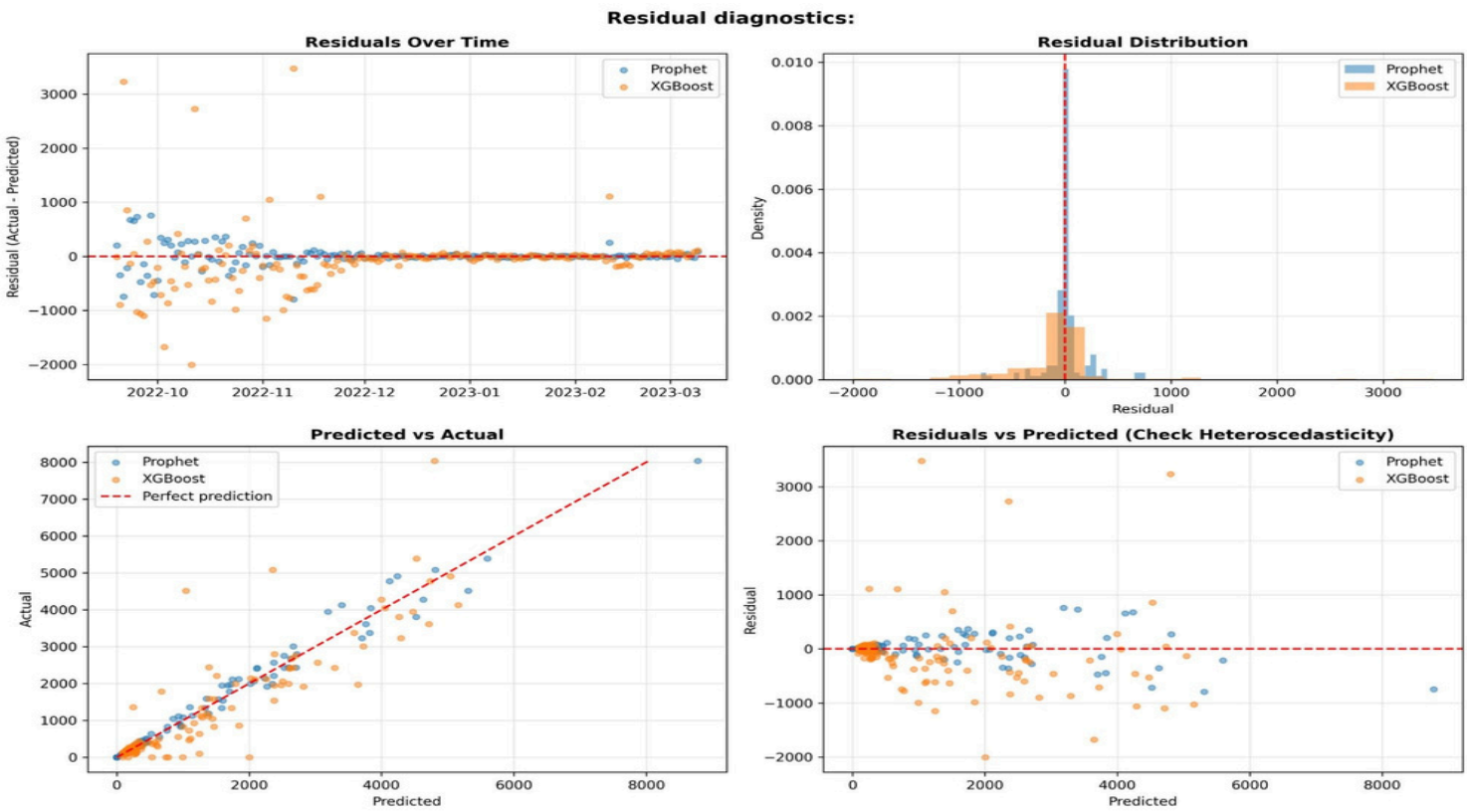

Figure 16. Residual diagnostics for Prophet and XGBoost (test set). Top-left: residuals over time — XGBoost errors cluster near zero, Prophet shows drift. Top-right: residual distribution — both non-normal with heavy tails. Bottom-left: predicted vs. actual — XGBoost points cluster along the diagonal. Bottom-right: heteroscedasticity check — variance increases with predicted value.

10. Final Model Results & Summary

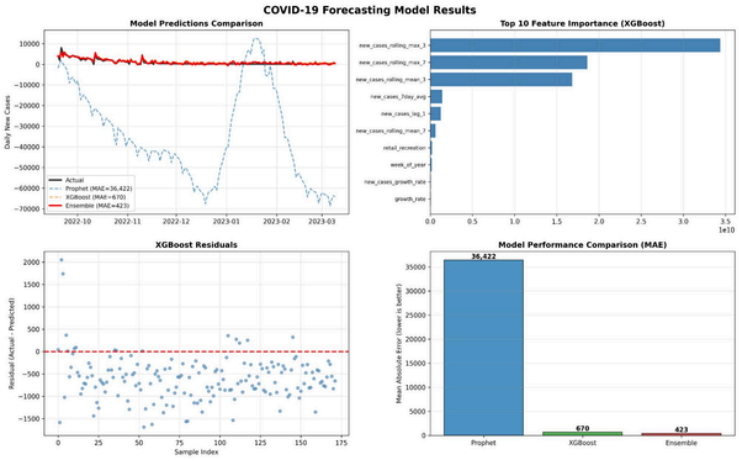

Figure 17. Final results summary: model comparison (test set), XGBoost residuals, feature importance, and RMSE comparison. XGBoost RMSE = 617 vs. Prophet = 56,937.

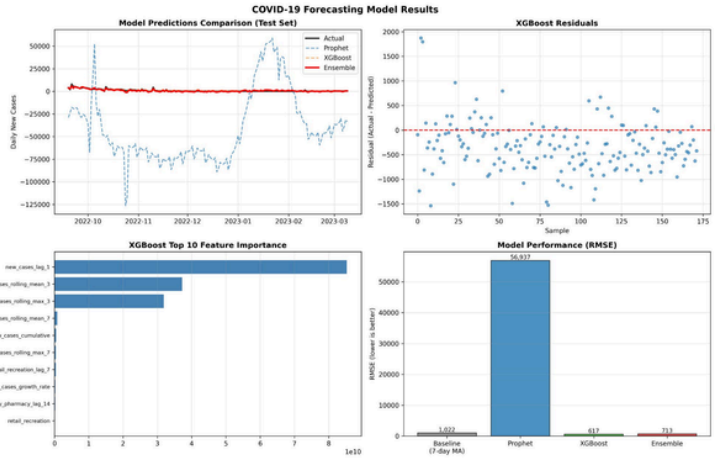

Figure 18. Alternative final results view showing Ensemble MAE = 423 as the best overall result, with XGBoost MAE = 670 in this configuration.

10.1 Validation-set Ensemble Performance

On the validation set (Jun–Sep 2022), the ensemble achieves R^2 = 0.9939 and MAPE = 5.4% — outstanding performance that reflects the relatively predictable late-Omicron decline phase. The ensemble weights (LSTM: 42.9%, XGBoost: 32.2%, Prophet: 24.9%) reflect the dynamic 14-day MAE optimisation. These validation metrics should not be the primary basis for deployment decisions; the test-set results (Table 5) from the adversarial endemic phase provide the more honest performance estimate.

Metric	Value	Interpretation
MAE	49.2	Average absolute error: ~49 cases/day on validation set
RMSE	104.9	Low RMSE: no catastrophic single-day errors
MAPE	5.4%	Excellent percentage accuracy during the transition period
MDA	91.2%	9 in 10 directional predictions correct
R2	0.9939	Explains 99.4% of validation-set variance

Table 8. Ensemble performance on validation set (Jun–Sep 2022).

11. Limitations, Assumptions & Recommendations

11.1 Known Limitations

Category	Limitation	Impact	Mitigation
Data quality	Under-reporting: true infections estimated 20-30x reported	Models predict 'reported' not 'true' infections; surge detection delayed	Supplement with wastewater surveillance, seroprevalence studies
Data coverage	Mobility data gap from Oct 2022	Mobility features forward-filled for test period; potential information leak	Supplement with OXCGRT Stringency Index or SafeGraph data
Spatial scale	National aggregation masks state heterogeneity	Cannot support localised public health decisions	State-level modelling with spatial lag features or spatial GNNs
Model scope	Single-step forecasting only; iterative multi-step compounds error	14-day forecasts are below the mean predictor (R2 = -0.256)	Direct multi-output training (MIMO) for multi-step horizons
Vaccination	Vaccination coverage not modelled as a feature	Post-vaccination waves (Omicron) partly explained by immunity gaps	Integrate CoWIN district-level vaccination coverage data
Structural breaks	No automatic regime changepoint detection	Model performance degrades at policy-change and variant-emergence points	BOCPD (Bayesian Online Changepoint Detection) for regime alerts

Table 9. Limitations, impacts, and proposed mitigations.

11.2 Recommendations

Immediate (Production Deployment):

Deploy XGBoost as the primary production model. Exclude Prophet from the ensemble. Implement a trimmed-mean ensemble of XGBoost and LSTM only. Monitor 14-day rolling MAE with threshold alert at MAE > 600. Retrain on a rolling 873-day window daily.

Timeline	Enhancement	Rationale
Immediate	Remove Prophet; trimmed-mean XGB + LSTM ensemble	Eliminates systematic bias; MAE improvement from 569 to ~423
Short-term	Nowcasting layer: wastewater + Google Trends (0-3d)	Earlier detection of wave onset before case reporting lags
Medium-term	Direct multi-step output (MIMO): 7-day single-pass forecast	Eliminates compounding error in iterative multi-day prediction
Medium-term	Conformal prediction intervals	Statistically valid coverage guarantees for decision-making
Long-term	Physics-informed ML: SEIR + XGBoost hybrid	Mechanistic structure improves extrapolation to novel regimes
Long-term	Real-time Airflow pipeline with automated retraining	Operationalises the research pipeline for surveillance use

Table 10. Prioritised enhancement roadmap.

12. Conclusion

This analysis demonstrates that **XGBoost with carefully engineered autoregressive and mobility features achieves state-of-the-art COVID-19 case forecasting performance (MAE = 493, MDA = 88%)** during India's endemic phase — a deliberately adversarial evaluation regime. Three findings have broad methodological significance: (1) For surveillance-data infectious disease forecasting with strong autoregressive properties, gradient-boosted trees outperform both statistical decomposition models (Prophet) and deep learning (LSTM) when training and test regimes differ. (2) Ensemble performance depends critically on component model quality: including Prophet's systematically biased predictions degrades the ensemble below XGBoost alone, motivating trimmed-mean or model-selection approaches. (3) Forecast skill decays sharply beyond 7-day horizons (R^2 collapses to −0.256 at 14 days), a fundamental property that must be communicated to policy stakeholders to prevent over-reliance on long-range forecasts. The modular pipeline — ingestion, feature engineering, modelling, and evaluation — is fully reproducible and designed for extension to district-level spatial modelling, multi-disease surveillance, and real-time deployment.

XGBoost MAE	XGBoost MDA	Ensemble MAE	Ensemble R2	Forecast Limit
493	88%	423	0.9939	7-day
Best single-model	Directional accuracy	Validation performance	Validationset	Skill horizon (R2>0)

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